BANTAYBAHA: FLOOD MONITORING SYSTEM WITH SMS NOTIFICATION

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Abstract

The advancements in technology have resulted to a new form of monitoring the flood, the use of Internet of Things. Internet of Things(IoT) can now connect every device imaginable with the right devices and software. It can be a person with a heart monitor implant, a farm animal, an automobile that has built-in sensors to alert the driver when tire pressure is low, a device that checks the volume of water in the farm through the use of sensors and many more. The researchers chose "BantayBaha: Flood Monitoring System with SMS Notification" to assist on a barangay called Virgen Delas Flores which is located in Baliwag, Bulacan. The method used by the barangay officials to monitor the flood is by using measurement bars on their posts. Integrating the Flood Monitoring System to their barangay would really help them be more prepared and lessen their burden because the device itself will send an SMS notification to the residents of the barangay indicating the water level of the flood and the data will be recorded and saved to the website .a realistic perspective of the university campus.

Keywords: Internet of Things, SMS notification, monitoring system, flood

In many parts of the Philippines, floods make harms on properties and it brings an enormous misfortune to people and governments. The researchers conducted a research on the barangays located in Baliwag, Bulacan and decided to choose the nearest and most frequently flooded barangay which is the barangay Virgen Delas Flores. The method used by the barangay officials to monitor the flood is by putting a measurement bar on the posts in their area . The barangay officials use a megaphone to announce the water level of flood in their barangay.

As stated in the study of Souza A., Curvello A. and Souza F. et al in a flood warning system to critical region (2017), "In light of the E-noe 6 project, this paper is created to decrease harms and misfortunes brought about by flood in urban focuses, through the modest flood cautioning framework to authorities and population. The framework screens the water level of a runway or stream and as per the hazard with the level, triggers crisis alarms to the authorities through phone calls and SMS messages. For the population, there are light guides that characterize the circumstance of the stream. The information is gathered in distributed storage through the Ubidots server, going conceivable to picture them progressively".

The researchers will be using Internet of Things (IoT) which is a system of interrelated computing devices, mechanical and digital machines, objects, animals or people that are provided with unique identifiers and the ability to transfer data over a network without requiring human-to-human or human-to computer interaction. Thus, in this project, the incorporation of IoT, to make a simple flood monitoring system into a smart flood monitoring system will be sending SMS notification to the people so that they can evacuate and can prevent destruction of properties due to flood. With the use of SMS notification enhanced to this flood monitoring and warning system, it will greatly satisfy the needs of the people especially the government. It will make the people more alerted on the situation.

As mentioned by Whitmore A., Agarwal A. and Xu L. in The Internet of Things: a survey of topics and trends (2015), "The Internet of Things is a worldview where ordinary articles can be outfitted with recognizing, detecting, systems administration and preparing capacities that will enable them to speak with each other and with different gadgets and administrations over the Internet to achieve some target. Eventually, IoT gadgets will be pervasive, setting mindful and will empower surrounding insight. This article covers the ebb and flow condition of research on the Internet of Things by inspecting the writing, distinguishing momentum patterns, depicting difficulties that compromise IoT dissemination, exhibiting open research questions and future bearings and gathering an extensive reference rundown to help analysts".

Moreover, flood monitoring is one of the critical tasks to do to ensure the safety of the people. Especially whenever there is a strong typhoon that will hit the country.

The purpose of this project is to prevent the feeling of being anxious by the flooding caused by the strong typhoon or heavy rain. In fact, the proposed project, BantayBaha, automatically sends SMS notification to the people whenever the flood water level is increasing, so they can be alerted as soon as possible to evacuate.

The device can create a logsheet that contains the water level, time, and phone number who received the text message. This logsheet will be displayed in the website. Eventually, the project mainly focuses on community where there is a possibility to get flooded such as Brgy. Virgen Delas Flores of Baliwag, Bulacan. Their streets are prone to flooding so this project will help them act right away whenever the flood water level is increasing to prevent further damages. They could expect the substantial benefits from this project because of the convenience it can actually give.

Project Context

The project seeks to integrate an innovative flood monitoring system with SMS notification. The proponents proposed a flood monitoring device that will monitor the water level of the place and when it reaches a certain level it will automatically send a text message using SMS notification to the people to alert them. The main objective of this study is to introduce a flood monitoring system, to help the citizens be prepared and knowledgeable whenever there is flood. 1.3 PURPOSE AND DESCRIPTION OF THE PROJECT The project proposed by the researchers is an Flood Monitoring System with SMS notification for Virgen Delas Flores (VDF). The project also includes user management system. The Internet of Things allows the users to monitor the flood online. Moreover, it is fully customized according to the needs of the barangay which also means that more features can be added or be modified in the future.

Objectives of the Project

The main objective of this project is to create a technologically advanced flood monitoring device that can send SMS notification to the people regarding the water level status of their barangay. Specific Objectives: The following are the objectives of BantayBaha: Flood Monitoring System: 1. To monitor the water level of the flood. 2. To send an SMS notification update to the residents of Virgen Delas Flores (VDF), Baliwag Bulacan. 3. To inform the residence of VDF for evacuation in case it reaches the critical level using the LED light signals. 4. To have the capability to log the changes in water level. 5. To be able to add and delete users from the database.

Scope and Limitation

The BantayBaha: flood monitoring device can detect the water level of the flood. After detecting the water level of the area it will automatically send sms notification to the

residents of Brgy. Virgen Delas Flores, Baliwag, Bulacan. The device can create a logsheet that contains the water level, time, and phone number who received the text message. This logsheet will be displayed in the website. There will be a user interface to add or remove users from the database. This project does not use solar power supply as energy source. The system cannot determine the temperature of the flood. The device cannot control or remove the flood of the area affected.

Significance of the Project

Residents of Brgy. Virgen Delas Flores, Baliwag, Bulacan This project will help the residents to be informed of the upcoming flood and to prepare for evacuation if necessary. Barangay Officials This project will help the barangay officials to be more cautious and alerted in terms of monitoring the flood level of their barangay. Schools This project will help the schools to react or suspend the classes immediately whenever they receive alert messages from the flood monitoring device.

Review of Related Literature and Studies

Related Literature

Based on the article of Lee K. (2015), The Internet of Things (IoT), also called the Internet of Everything or the Industrial Internet, is an innovation model that is equipped for interfacing with one another with the utilization of a worldwide system of machines and gadgets. The IoT is getting the consideration from the wide scope of enterprises remembering it as one of the significant territories of future innovation.

Based on the article of Lee K. (2015), Internet of Things is also called Internet of Everything since these days, everything that can be associated can interface with the web. Internet of Things is currently being gazed upward to get in

the IT business. Utilizing various machines and gadgets that can be associated with the web is proficient to relate with one another. Individuals admire this as their future in the innovation world that will make their lives simpler and will make them increasingly beneficial, in corporate ventures also in the family units of each network.

As said by Madakam and Tripathi et al. (2015), the arrival of IT and ITes technologies make a colossal effect on the association's working conditions in people's everyday lives. It is being a verifiable truth crosswise over business sectors on account of its few applications incorporating men's regular day to day existence in the general public. The improvement of the Internet of Things (IoT) has been essentially determined as a result of the necessities of enormous companies that incredibly advantage from the prescience and consistency given by the capacity to pursue all items that are put through the procedure utilized by firms to accumulate assets, change them into merchandise or item, and appropriate them to the purchasers.

As said by Madakam and Tripathi et al. (2015), Human life is changing regularly as various innovation in the IT business ceaselessly comes through. It makes a major effect on everyone's regular daily existence as it is being applied in nearly everything occasionally, regardless of whether it is on an association or on one's family. It is in effect surely understood known as a result of the significant advantages it can give extraordinarily for those in enormous partnerships that have huge necessities of innovation use. It enables organizations to be increasingly proficient, and to have a progressively adaptable authoritative framework.

As stated in the article of Mahar A., Lagmay F. and Racoma B. et al. (2017) The Philippines being a locus of tropical typhoons, tidal waves, seismic tremors and volcanic emissions, is a hotbed of calamities. These common dangers cause loss of lives and expensive harm to property. Arranged in an area where atmosphere and geophysical storm is

normal, the Philippines will definitely experience the ill effects of disasters like those accomplished as of late. Proceeded with improvement and populace development in peril inclined regions, it is relied upon that harm to foundation and human misfortunes would endure and even ascent except if proper measures are promptly actualized by government.

According to Kundzewicz Z., Kanae S. and Seneviratne S. et al. (2014) An all-encompassing point of view on evolving precipitation driven flood hazard is accommodated during the late twentieth and mid-21st hundreds of years. Financial misfortunes from floods have enormously expanded, chiefly determined by the growing presentation of benefits in danger. It has not been conceivable to characteristic downpour created top stream flow patterns to anthropogenic environmental change in the course of recent decades. Anticipated increments in the recurrence and power of substantial precipitation, in light of atmosphere models, ought to add to increments in precipitation-created neighborhood flooding (for example blaze flooding and urban flooding).

As stated by Brooks G. and George S. (2015) Significant flooding impacts networks and country zones, including the City of Winnipeg, had caused critical flood debacles in the nineteenth and twentieth hundreds of years. Since 1950, a variety of basic flood control measures had been developed (and some later moved up) to relieve flooding, including two preoccupation channels, a flood control dam, dyking (straight and ring dyking), and had raised earthen cushions under structures. As said by Che D. and May L. (2017) The use of a streamlining/reenactment model for the mimicked continuous flood control for waterway repository frameworks to the cataclysmic May 2010 flood on the Cumberland River at Nashville, Tennessee is depicted. The advancement/reenactment model incorporates five significant parts, including a hydrologic precipitation spillover model, a water driven temperamental

stream model, a momentary precipitation estimating model, a supply activity model, and a hereditary calculation improvement model.

As stated in the article of Rahman M., Rahman A. and Chowdhury S. (2017) This examination has utilized a survey of the existence cycle of flood moderation extends, an audit of manageability evaluation procedures, meetings with specialists and contextual investigations including two flood alleviation extends in Australia.

According to Ghanbarpour M.R , Saravi M. and Salimi S. (2014) The outcomes have demonstrated that expressed WTPs essentially fluctuates with family unit pay, separation individuals live from the stream and the land use sort of properties. Discoveries of this investigation propose that most of respondents view flood peril as the most significant cataclysmic event. Moreover, WTPs are altogether higher for the individuals who have elevated level of flood hazard recognition.

As mentioned by Prawiranegara M. (2014) The Philippines is one of the most normal peril inclined nations on the planet, particularly for water-related calamities (floods and avalanches) activated by storm. This investigation is spurred by the outrageous flooding occurrence in Metro Manila and encompassing zones in September 2009 during the most obliterating hurricane to hit the nation named as Tropical Storm Ondoy. Exceptionally urbanized urban communities of Metro Manila in the downstream of Marikina River Basin endured the most and the debasement of the upper piece of the bowl in Rizal Province has been distinguished as one of the fundamental driver of such huge flooding.

As stated by Beven K. and Hall J. (2014) Flood chance administration is a procedure of basic leadership under vulnerability. Customary ways to deal with dissecting floods accepted a pretense of determinism however ideas, for

example, "structure floods", and dealt with vulnerabilities verifiably and hazily through freeboard remittances and different variables of wellbeing.

According to Tahir W., Jani J. and Endut I. et al. (2016) Flood is a characteristic climate related fiasco as often as possible happening in Malaysia. Perhaps the best challenge that Malaysia faces today is perceiving the greatness of dangers presented by flooding. The general population, private and NGOs should ponder the measure of speculations required to lessen the flood chance, including making proper crisis arrangements, fortifying the current Standard Operating Procedures (SOPs), and finding new answer for limiting danger identified with flood catastrophe. The current existing SOPs show that there is still absence of all-encompassing flood chance administration framework to limit this issue. The nation ought to be resolved to set up a national approach on flood chance administration that requires viable, affordable, reasonable, and steady administration of flood hazard to individuals, properties, and networks

Methodology

Requirement Analysis

This section includes the analysis of the information that the researchers have obtained through their observation and study. User Need Analysis. The project was not only for barangay officials but also for any concerned citizen and different establishments. It can be used as a mean of informing them remotely. The proposed objective of the project is to cater the concern of the barangay constituents. It is done by providing a system that will allow them to automatically receive information regarding water level in their barangay that is a must need during rainy seasons. The BantayBaha device will be beneficial to the people who are affected. Methods to create the project. The researchers first researched about flood and how to monitor floods then the

researchers conducted an interview on the head rescue team of brgy. Virgen Delas Flores, Baliwag, Bulacan to get information from him about the flood in their barangay. The researchers made a list of the parts of the device that will be needed for the project.

Requirements Documentation

This section includes the sources that are needed for the documentation that are acquired by the researchers through data gathering

Source of Data

The researchers concluded an interview with barangay officials about the corresponding water level that will indicate which level is normal to critical. The researchers also made the project progressive through the use of the following sources of data: Internet. The researchers were able to obtain information on the scholarly articles through browsing the internet. The researchers utilized educational websites and e-books for further information about the project.

Responsibilities

Capstone Project Leader Leads the members to find new heights in conducting the research project. The one who manages and assigns tasks for each member. Has the ability to take leadership of the work that the member cannot do. The leader is also the person who keeps the team together and organizes the needed materials. Capstone Documentation Writer. Conducts the process and requirements of the project. Collects the information used for the project. The person who does all the paper work about the project. The documentation has different chapters and includes Introduction, Related Literatures and Studies, Technical Background, Methodology, Results and Discussion, Recommendation, References and Appendices. Capstone

Program Designer. The person who designs and codes the program to meet the needs of the project.

Verification, Validation and Testing

For the verification the researchers made it certain that the design specifications requirements and necessary functionalities of the web interface and the system are attained. For the validation the researchers performed code inspections, regular repetitions of testing and simulation of the device. For the testing the researchers carried out various types of verification and validation to try the overall quality of the entire system to steadily avoid and point out the defects, errors, missing requirements and risks.

Results and Discussion

The researchers conducted an evaluation for the proposed project. This chapter consists of results regarding the assessed evaluation. With the help of Baliuag University students the proponents evaluated 70 respondents with the given criteria to test the website and to know if the proponents meet their objectives

Every criterion below is scaled from numbers 1 to 5. With 5 as the highest scale, it is rated as “Excellent”, number 4 rated as “Very Good”, number 3 rated as “Good”, number 2 rated as “Fair”, and number 1 rated as “Poor”.

I. System Functionality

	Question	Indicators					
		E	VS	S	P	B	Mean
1.	The device can be used to remotely monitor the flood.	54	16				4.77
2.	The device can send alert notifications to the phone.	46	24				4.65

Continuation

	Question	Indicators					
		E	VS	S	P	B	Mean
3.	The float switches successfully detects water.	50	19	1			4.69
4.	The LED lights turn on/off which correspond to the water level.	48	20	2			4.65
5.	The GSM module has a good signal reception.	55	15				4.78
Grand Mean							4.70

As a result of the evaluation conducted by the researchers about the system functionality, fifty-four respondents answered excellent that the device can be used to remotely monitor the flood via internet and there are sixteen respondents who were very satisfied on it. Forty-six respondents likewise answered excellent that the device can send alert notifications. There are twenty-four respondents who were very satisfied on it. Fifty respondents answered excellent that the float switches successfully detects water. There are nineteen respondents who are very satisfied on it and one respondent was satisfied. Forty-eight respondents answered excellent that the LED lights turn on/off which correspond to the water level. There are twenty respondents who were very satisfied on it and two respondents were satisfied. Fifty-five respondents answered excellent that the GSM module has a good signal reception. There are fifteen respondents who were very satisfied on it.

II. System User Interface (UI)

	Question	Indicators					
		E	VS	S	P	B	Mean
1.	The web interface is easy to navigate.	33	31	6			4.38
2.	The web server in phone is easy to use.	44	25	1			4.61

Continuation

Question	Indicators					
	E	VS	S	P	B	Mean
3. The web server is user-friendly.	40	29	1			4.55
4. The design of the website is appealing.	40	25	5			4.49
Grand Mean						4.50

When it comes to the user interface. Thirty-three respondents answered excellent and found the web server easy to navigate, On the other hand, thirty-one respondents chose to answer very satisfactory and six respondents answered satisfactory. Forty-four respondents answered excellent and found the web server in phone easy to use. On the other hand, twenty-five respondents were very satisfied and one respondent is satisfied. Forty respondents answered excellent and found the web server user-friendly. On the other hand, twenty-nine respondents are very satisfied and one respondent is satisfied. Forty respondents answered excellent and found the website design appealing, On the other hand, twenty-five respondents chose to answer very satisfactory and five respondents answered satisfactory.

III. System Reliability

Question	Indicators					
	E	VS	S	P	B	Mean
1. The device can function in any environment.	38	27	5			4.47
2. The device is user intuitive.	35	35				4.5
3. The device can be easily maintained.	40	28	2			4.54
4. The device can help the residents.	54	15	1			4.75
5. The parts of the device are durable.	43	24	2			4.58

Continuation

Question	Indicators					
	E	VS	S	P	B	Mean
6. The device can be controlled easily.	46	22	2			4.62
Grand Mean						4.57

When it comes to the system reliability. Thirty-eight respondents answered excellent that the device can function in any environment and there are twentyseven respondents who were very satisfied and five respondents were satisfied. Thirty-five respondents answered excellent that the device is user intuitive and thirty-five respondents who are very satisfied. Forty respondents answered excellent that the device can be easily maintained. There are twenty-eight respondents who were very satisfied and two respondents were satisfied. Fiftyfour respondents answered excellent that the device can help the residents. There are fifteen respondents who were very satisfied and one respondent was satisfied. Forty-three respondents answered excellent that the parts of the device are durable. There are twenty-five respondents who were very satisfied and two respondents were satisfied. Forty-six respondents answered excellent that the device can be easily controlled. There are twenty-two respondents who answered very satisfactory and two respondents answered satisfactory. The researchers conclude that most of the respondents are very satisfied with the functionality of the system and gave it an average of 4.70 (Very Satisfactory). Most of the respondents are also very satisfied with its user interface and gave it an average of 4.50 (Very Satisfactory) and also agreed that the system is reliable so they gave it an average of 4.57(Very Satisfactory). The following chapter concludes this project. Likewise in this chapter, a summary of the project is presented and findings of the project will be discussed and interpreted. Moreover, the significance of this project to the Internet of Things will be analyzed. These conclusions might be held inaccurate,

although these conclusions are relevant to the advancement of Internet of Things in this modern age.

Conclusion

The conclusion that can be taken from the project is that having a device like BantayBaha that can monitor flood water levels is a great help especially in a country like Philippines where due to storm and typhoons running rampant all year long. The conclusions from this project are not statements such as “Having a device such as this is a must in every barangay in the nation.” Instead, results from qualitative and quantitative methodologies are presented with the implications that this project should be tried by others and studied more.

Recommendations

The researchers would like to recommend future researchers to improve the BantayBaha device by taking different approach such as visual studio based interface where instead of HTML format they can use an .exe file type. The researchers would like to recommend future researchers to improve the BantayBaha device by adding peripherals such as a solar power supply to make the device self-sufficient. For the instance where the location you want to place a BantayBaha have drastic temperature discrepancy adding a module that can determine the water temperature would be beneficial. Future researchers are allowed to use and to improve the BantayBaha device given that it will improve the features of the system

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